theories and methods were introduced, naming a few are Probability Theory, Sampling Theory, Statistical Inference, Design of Experiments, Correlation and Regression Methods, Time Series and Forecasting Techniques.

In early 1900s, statistics and statisticians were not given much importance but over the years due to advancement of technology it had its wider scope and gained attention in all fields of science and management. We also tend to think statistician as a small profession but a steady growth in the last century is impressive. It is pertinent to note that the continued growth of statistics is closely associated with information technology. As a result several new inter- disciplines have emerged. They are Data Mining, Data Warehousing, Geographic Information System, Artificial Intelligence etc. Now-a-days, statistics can be applied in hardcore technological spheres such as Bioinformatics, Signal processing, Telecommunications, Engineering, Medicine, Crimes, Ecology, etc.

Today's business managers need to learn how analytics can help them make better decisions that can generate better business outcomes. They need to have an understanding of the statistical concepts that can help analyze and simplify the flood of data around them. They should be able to leverage analytical techniques like decision trees, regression analysis, clustering and association to improve business processes.

1.2 Definitions

Statistics has been defined by various statisticians.

- *'Statistics is the science of counting'* -***A. L .Bowley***
- *'Statistics is the science which deals with the collection, presentation, analysis and interpretation of numerical data'* - ***Croxton and Cowden***
- **Wallist** and **Roberts** defines statistics as *"Statistics is a body of methods for making decisions in the face of uncertainty"*
- **Ya-Lun-Chou** slightly modifies Wallist and Roberts definition and come with the following definition : *"Statistics is a method of decision making in the face of uncertainty on the basis of numerical data and calculated risk."*

It may be seen that most of the above definitions of statistics are restricted to numerical measurements of facts and figures of a state. But modern thinkers like Secrist defines statistics as

'By statistics we mean the aggregate of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated to reasonable standards of accuracy collected in a systematic manner for a predetermined purpose and placed in relation to each other'.

Ch1_11th Stats.indd 3 2/3/2019 10:56:29 AM